02/06/26 Brightlearn

Data analysis

1. #Databricks *
   - Get a certificate
   - take exam
   - (Alex JT) for beginners
   - Alex the Analyst
2. Microsoft credentials (certificate)
   - +Rico to write an exam
3. GitHub (to build a portfolio, store codes, documentation, Presentation)
   - GitHub signup - on Google
4. CANVA (to build presentation)
5. Ekel (or google sheet)
6. AI (AI tool, Viewing dashboard)
   - Data studio
   - Google cloud (video) AI)

JT-corbin

GitHub – repository (where codes are VS stored in CLOUD)
(Public or Private)
branches – MAIN
(duplicate NewFolder)

- If you make mistakes on branch – you can always go back to MAIN
- If branch code is good, you can merge with main & delete the branch (not needed anymore)

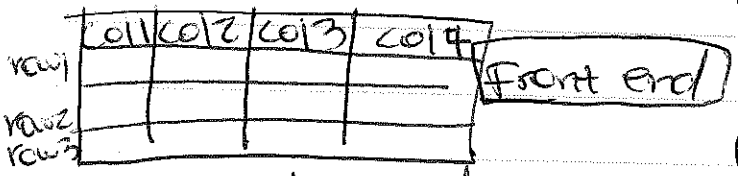
Green means – Information added
Red – Information removed

GitHub Desktop

03/06/26

Data
- Collection of facts
- Is information

Structured Data
- data stored in a table with rows & columns
- Consistent format



- e.g Spreadsheet
  - (Excel file)
  - membership Form
  - Front end
- Back end

Unstructured
- No specific format
- Inconsistent format
  - e.g Music
  - Picture, video
  - Podcast
  - Social media Post
  - PDF, docs
  - website
  - YouTube

coding
Data cleaning
miro.com
loveable.dev
draw.io - diagrams
read up
info to databases - doc

data collection

Front-end

Name:
Surname:
Age:
Location:

back-end

Id	Name	Surname	Age	Location
1	Rofhiwa	Neny	22	Sthlb
2	Sfiso	Nxumalo	23	CPT
3	Brian	Zuma	20	DBN

Unique I.D
e.g Order number

Order_I.D	Product	Revenue	Datetime
101	Str 2	R40	data
102	Ice	R20	Collection
111	All Star Bar	R80	03/06/2026 13:00

Computer order
POS system (Point of Sale)
Connected to a database (server room)
transacting

#Lookalike audience - data makes you see adverts of things you previously bought.

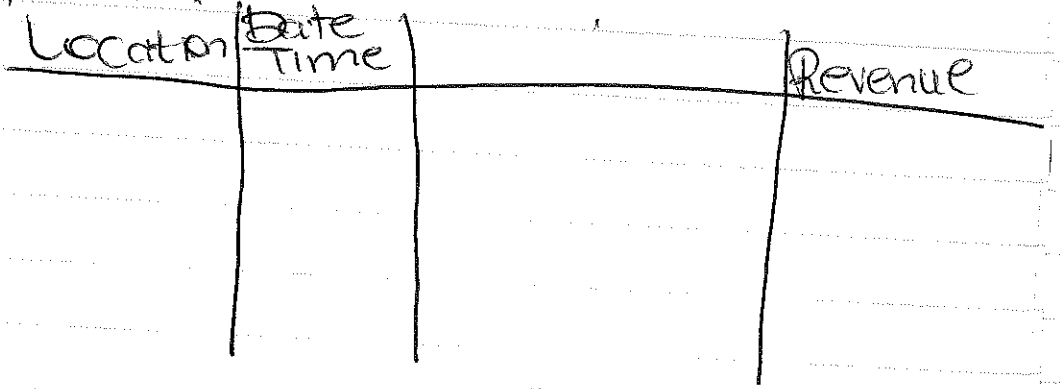
#To make data-based decisions

data gives more information about customers

#to make business decisions

Data is a time-Based

#Patterns



Cross-selling e.g combo
Upselling lotion 1-R40
2-R60
e.g body lotion + shower gel cheaper if you buy both.
e.g voice minutes + data combo

04/06/26

Data Flow

real time database that stores information

(OLTP)

Online Transactional Processing Database

Shoprite Tzaneen

DB1

(ETL) Extract Transport Load

(warehouse)

Shoprite Head Office

Shoprite Thohoyandou

DB2

ShopriteProtoria

DB2

(OLAP) Online analytical Processing

DWData warehouse

Process

Data analyst & data scientists operate from

does not allow/not used for analysing data, Used to capture data

- Create
- Insert
- Update
- DELETE

ETL Pipeline

transport data from the source to where it is consumed. (warehouse)

created by data engineer

#Data architector maps out how (Plan) the data works.
#Data Scientist - Predicts the future
#Data analyst - report on what happened yesterday.

Data warehouse - well organized library where data is stored.

ETL vs ELT

Databricks - data engineering - Ingestion - From Source (data collection) - Upload volumes & files OR Create/modify OR - Databricks connectors.

Data Management System
Eg memory stick - Data warehouse
data Soure

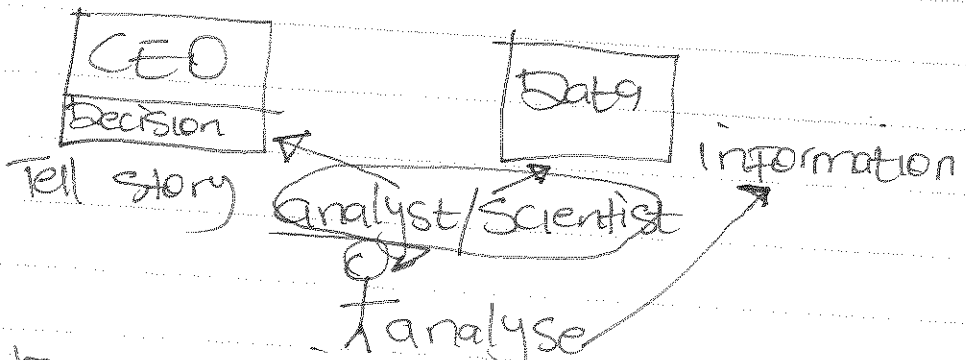
memory card - In order to use - need to put it in your phone.
Phone - DMS (Data Management System)

DMS - Interphase that gives us access to data

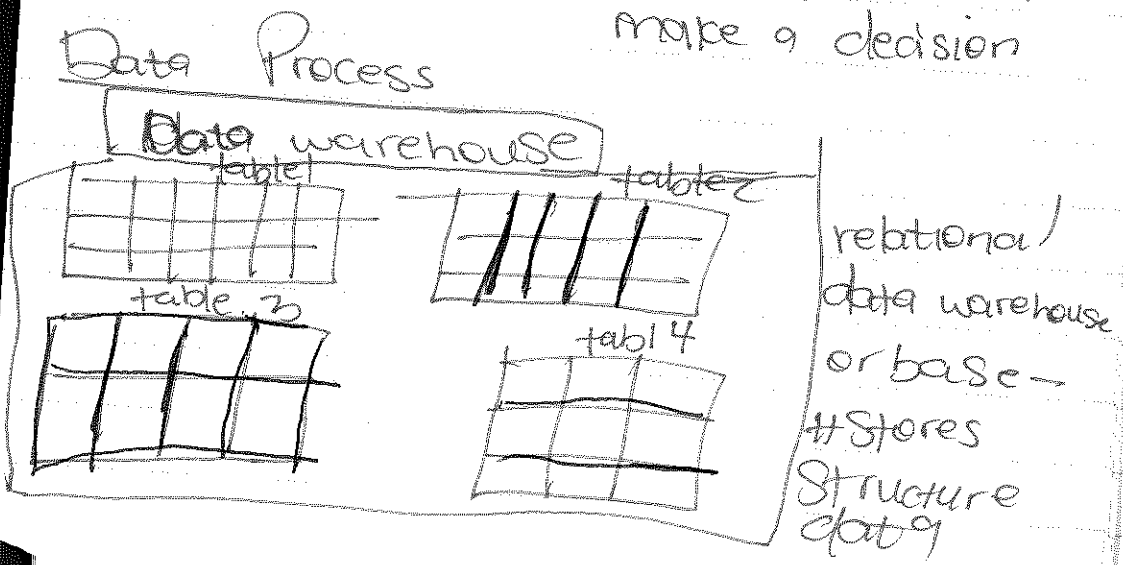
05/06/26

Data Visualization Bootcamp

Data → information
Give business decisions



Data Process



That can be related/joined together the become one.

Structured data (dimensions tables & fact tables) that can be used connected to one another using primary key & foreign key.

Dimension table
Customers

cust_ID	name	Age	City
1	Rafi	28	NJ
2	Jeff	22	LA
3	Mark	18	MG

Transactions

order_id	cust_id	Prod_ID	Rev
324	2	104	70
333	1	102	90

Dimension table
Products

Prod._ID	Prod_Category	Prod_name
101	Fruits	Banana
102	Fruits	Apple
103	Electronics	Laptop
104	Clothing	Uzzi Jeans

Dim table - contains discriptive information
Does not constantly change

Primary key - Unique identifier that differentiate each component from

E.g I.D number

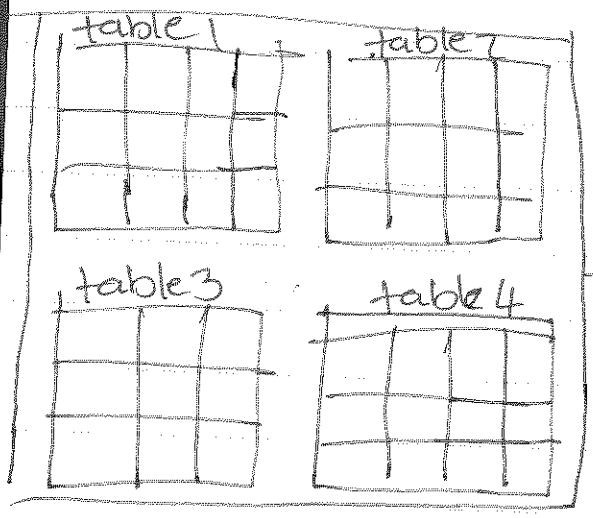
Transactions			
Order_ID	Cust ID	Prod I.D	Revenue
324	2	104	70
333	1	102	80
401	3	101	100

(fact table *
(Has Foreign keys
(Contains numbers
(Used for analysis

Use in conjuction with previous table. Customer name - represented by ~~Pro~~ Order number - tell a story stating what did the customer buy (Product I.D)

Foreign key - Primary key from one table that is found in another table

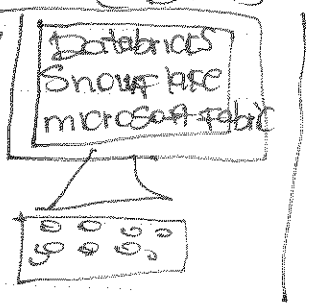
Data warehouse



Use DMS to access the data

Connected to our data

Data Management System (DMS)

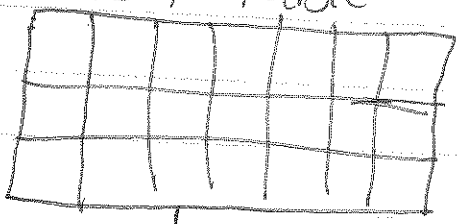


In a DMS we code (SQL, Python etc)

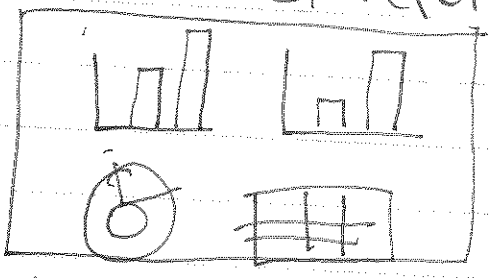
Why do we need to code?

- Data Inspection/exploration
  - To see what is available
- Data Exploratory Analysis
  - To understand the relationship btw, our data point.
- We code to retrieve, amend, update, manipulate, clean, aggregate, join, delete, update - with the goal of extracting actionable insights
- Feature building, creating features used in ML for predictions.

After coding (from previous 4 tables)
end. up with Summarized table
- Aggregated table
- Combined table
- Unified table
- Insight table



Power BI report



tell a data story, what do I want to show, kind of outcome? Revenue? Number of Customers.

visualization - Summarize complex table - extract insight

CML - machine Learning model

Python

How to choose which visualization or Graphs to use, when & How to use it and WHY

Before you ASK

Analyst ASK
DATA DRIVEN QUESTIONS

BRS - Business Requirements
The ASK - THE WHY REASON behind

CEO Want to understand venue for Coffe Shop for different Products

Understant what is needed by asking the right questions

to see
- would like daily revenue with VAT included or excluded?
- would you like to see repeat customer??
To analyze the frequency

Do you want the revenue normalized?
Focus specific categories

Data Processing
Code

Data analysis

Visualization
Create Graphs

Insights Generation

Data Presentation
Putting things together

Chpt 5 & 6

06/06/26

CHARTS - Comparisons & Trends

Pie chart

- Use when: You have 2-5 categories & the differences are easy to see

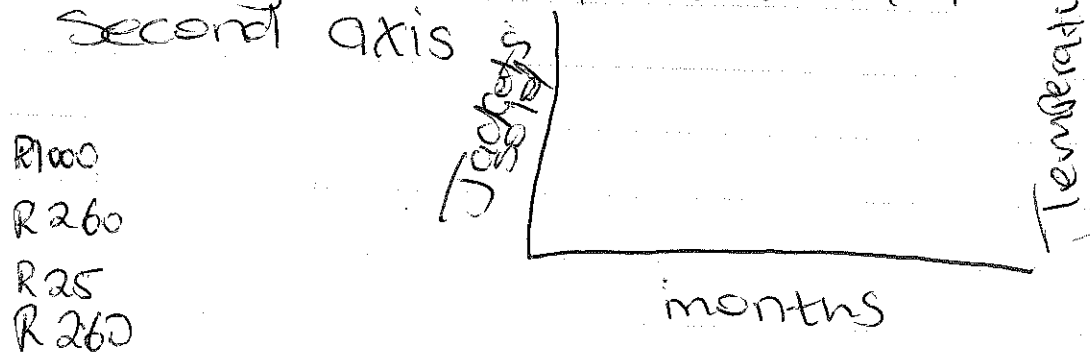
Donut chart

Pie chart with a hole — good for ~~do~~ dashboard summary cards.

2-4 categories

Combination chart

Two metrics on one chart — Usually colums + a line on a second axis

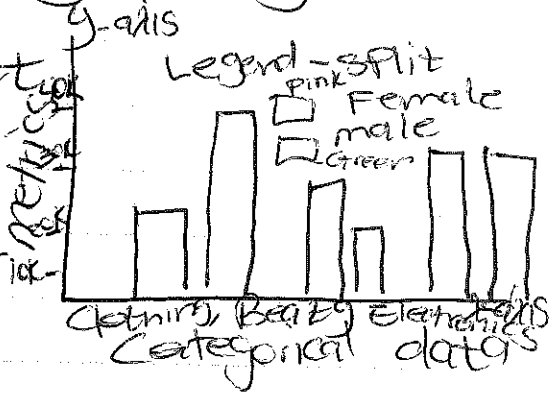


R1000
R260
R25
R260

Which type of Graph to use

The Data Design Blueprint
- choosing the right graph/chart for Effective Story telling

1. Bar Chart
Comparing Categories



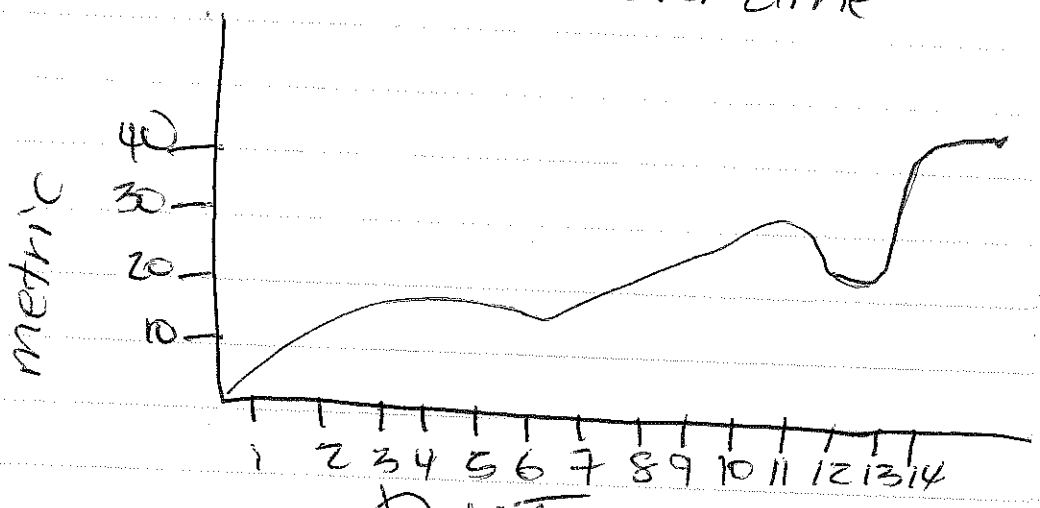
- Revenue
- no of customers
- No of Sales/ transactions

metric - refers to something we are measuring.
metric - Numerial data or quantitative (numeric)

- Categorical data - grouping
- mainly words
e.g Gender - M
- F

- A split or der legent gives you a detailed breakdown - Granularized

Line chart ~~showing~~ Trends over time

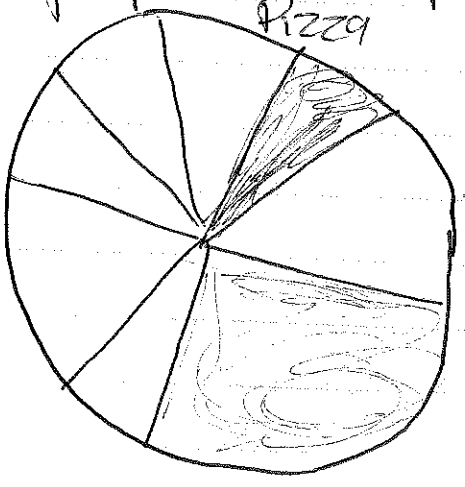


Date Time
- Month
- Day of week
- Year
- Date

} Continuous trends

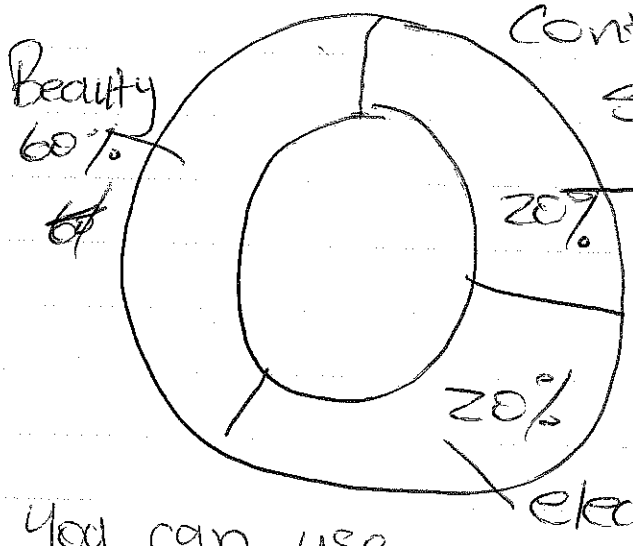
Pie chart

Showing ~~Proportions~~ Proportions



Categorical breakdown
- metric to proportion

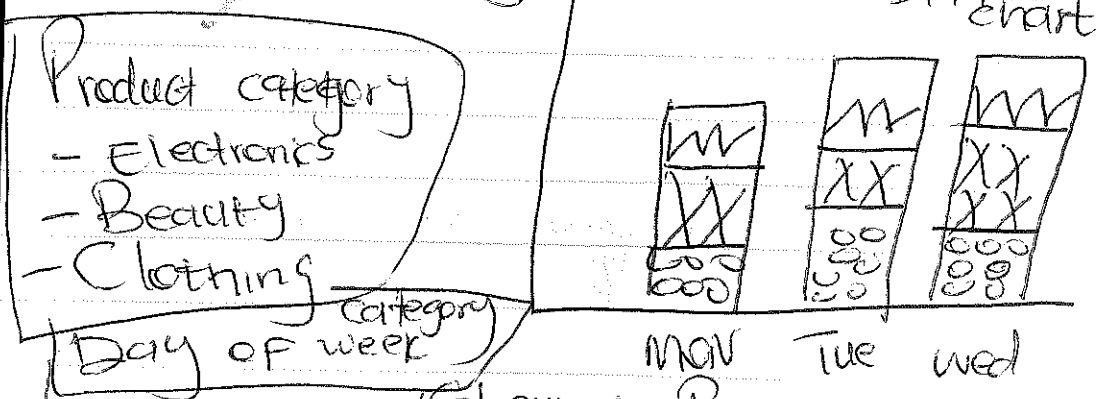
Show distribution
Contribution of something



Sum up to 100%
60+20+20 = 100%

You can use
stacked bar graphs to show proportions
OR even bar graph

stacked chart



Showing proportions for two different categorical data by specific metric

kaggle ***

Products
Time of day
Sales trends & time interval

which products generate the most revenue?

What time of day the store performs best?

Sales trends across products & time intervals

Recommendations for improving sales performance.

Excel

① Insert Pivot Table

Purchase date → Rows

Revenue per day → Values

~~Insert~~

→ click & drag-drop to rows

② New worksheet

Columns

Location stores

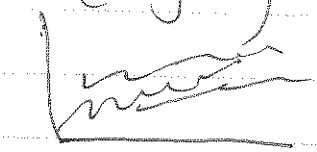
Filters

Pivot Table Fields

click on revenue (right)

Insert

Line chart

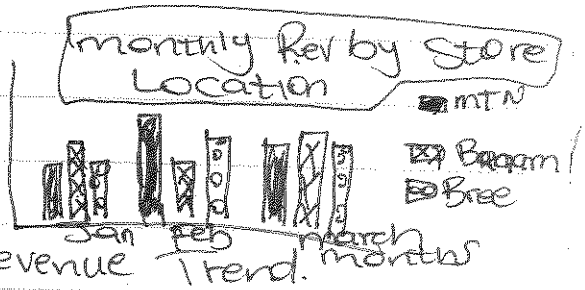


Filters:

Month

Design

Chart title: Revenue Trend.



E.g Rows: X-axis months names

Values: Put numerical data (metric

Columns: split

Legent: Store location

Bar chart

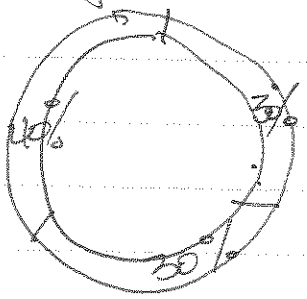
ON EXCEL - visualize

Get Processed data

Use Pivot Table on Excel

Show value as a percentage of total - right Pivot Table

- Pie chart
- Show data labels



Revenue distribution by store location:

Bree: 30%

MTN: 30%

Beam: 4%

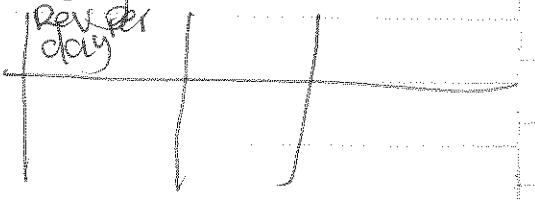
= 100%

Use same colours on different graphs - consistent colours!!

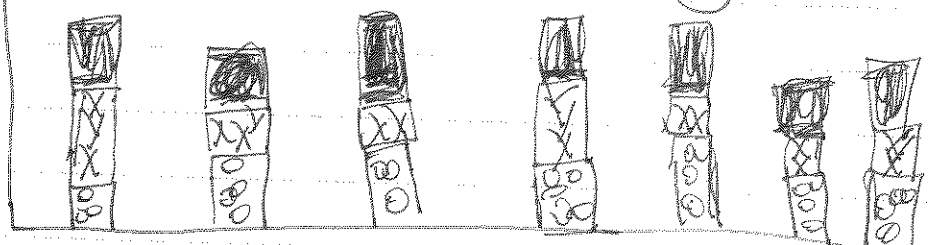
Values: revenue per day

Columns: store locations

Stacked Graph



Weekly Revenue split by Location



Lower Sun Mon Tue Wed Thu Fr Sat

Lower %

Hells %

Lower %

Astoria

Hells Kitchen

Lower

Data Studio

Create report

CSV File → create dataset

Coffee Bootcamps

Coffee shop analysis

upload → add

add chart

double tapp-top left of the screen to add the title e.g Coffee Shop

Add chart

→ create

Properties → to use in how you want your data to look like e.g type of chart

Data

Score Chart

Card

Setup

~~Data~~

Dimensions — categorical/ dates
e.g store location
3 stores
~~Donut~~ chart
Pie

metric:
no of stores,
revenue

& style — chart title
Bold
text size etc
colour font
Add boarder

2025/06/26 Introduction to SQL

SQL

Structured Query Language

↳ IS a scripting language that we use to interact with structured data

↳ we use SQL to retrieve information from a structured table. 'data'

Eg math test

students_results table

Student ID	name	marks
101	Rofhi	99
102	Mbali	25
103	Thabo	50
104	Lethabo	60

SELECT statements

↳ Allows us to select the specific columns from our table

2. The FROM Statement

↳ Allows us to specify the name of table we are selecting the data from

```
SELECT name, marks
FROM students_results
```

name	marks
Rofhi	99
Mbali	25
Thabo	50
Letnabo	60

write an SQL query to give me only the names of students that obtained more than 60% in exams

---

The select statement allows you to specify the columns you want in your final output
databricks - New Query

1. SELECT * - allows for ALL the columns from the table without specific column name one by one.
   - when you say SELECT * is to get all the column names in that table.

```
SELECT * Date, Gender, Age
FROM retail_sales_dataset;
```

When you are selecting specific

column names from a table, you seperate each column by a (,) comma

SELECT AGE,
Date,
~~Quantity~~ Quantity

(on the last column you specify, you do not put the comma)

SELECT DISTINCT ~~different~~ unique Statement

(we want to get different values from a specific column

(we remove duplicates/we do not repeat the same value

Syntax: SELECT DISTINCT
column name
FROM - table_name

Fruits orders

order Product_id	Products name	Price/revenue
101	Apple	R100
102	Orange	R50
103	Apple	R60
104	Banana	R150

which products are there?

SELECT DISTINCT prd_name
FROM fruit_orders

Prd_name
Apple
Orange
Banana

different values that base on specific column

SELECT Distinct Gender
FROM retail_sales ~~database~~ dataset